

# Random Forest Algorithm

## Simple Guide to Classification & Regression

Ananda Rimal

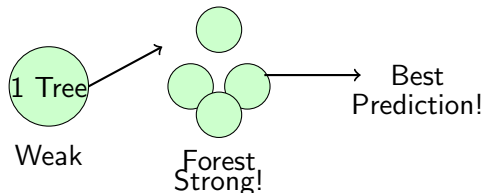
# What is Random Forest?

## Simple Definition

Random Forest = Many Decision Trees working together

### Think of it like:

- Asking 100 doctors instead of just 1
- Each doctor gives their opinion
- Final answer = Majority vote wins!



# Why Use Random Forest?

## Advantages:

- **High Accuracy:** Better than single tree
- **No Overfitting:** Many trees fix errors
- **Both Types:** Categories AND Numbers
- **Finds Important Features:** Knows what matters

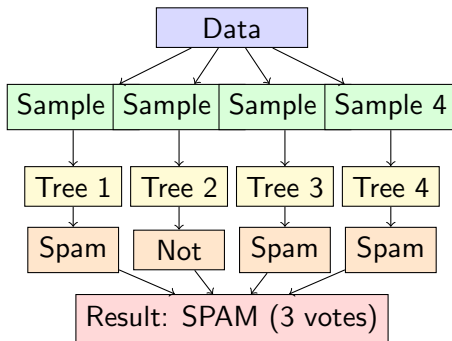
## Real Use:

### Example

- **Category Prediction:**
  - Email: Spam or Not Spam
  - Disease: Sick or Healthy
  - Customer: Will Buy or Won't Buy
- **Number Prediction:**
  - House price: \$300,000
  - Temperature: 25°C

# How Random Forest Works

- 1 Take Original Data
- 2 Create Random Samples
- 3 Build Tree on Each Sample
- 4 Each Tree Votes
- 5 Majority Wins!



## Example 1: Will Customer Buy?

**Dataset:** 10 customers with their details

| ID | Age    | Income | Student? | Buy? |
|----|--------|--------|----------|------|
| 1  | Young  | Low    | No       | No   |
| 2  | Young  | Low    | Yes      | No   |
| 3  | Middle | Low    | No       | Yes  |
| 4  | Senior | Medium | No       | Yes  |
| 5  | Senior | High   | Yes      | Yes  |
| 6  | Senior | Medium | Yes      | Yes  |
| 7  | Middle | High   | Yes      | Yes  |
| 8  | Young  | Medium | No       | No   |
| 9  | Young  | High   | Yes      | Yes  |
| 10 | Senior | Medium | Yes      | Yes  |

**New Person:** Age=Young, Income=Medium, Student=No

**Question:** Will this person buy? Yes or No?

# How Forest Decides - Customer Purchase

**New Person:** Young, Medium Income, Not Student

**5 Different Trees Give Their Opinion:**

| Tree   | Looks at             | Vote |
|--------|----------------------|------|
| Tree 1 | Age, Income          | No   |
| Tree 2 | Age, Student         | No   |
| Tree 3 | Income, Student      | No   |
| Tree 4 | Age, Income, Student | No   |
| Tree 5 | Age, Income          | Yes  |

**Vote Count:** No = 4, Yes = 1

**Final Decision**

Majority says **NO** → Will NOT buy!

## Example 2: Email Spam Detection

**Dataset:** 8 emails from inbox

| ID | Has "FREE" | Has "Win" | Many Links | Spam?    |
|----|------------|-----------|------------|----------|
| 1  | Yes        | Yes       | Yes        | Spam     |
| 2  | No         | No        | No         | Not Spam |
| 3  | Yes        | No        | Yes        | Spam     |
| 4  | No         | No        | No         | Not Spam |
| 5  | Yes        | Yes       | No         | Spam     |
| 6  | No         | No        | Yes        | Not Spam |
| 7  | Yes        | No        | No         | Not Spam |
| 8  | No         | Yes       | Yes        | Spam     |

**New Email:** Has "FREE", No "Win", Many Links

**7 Trees Vote:**

Spam, Spam, Spam, Not Spam, Spam, Spam, Not Spam

**Result: SPAM (5 out of 7 votes)**

## Example 3: Disease Diagnosis

**Dataset:** Patient symptoms and diagnosis

| ID | Fever | Cough | Fatigue | Sick? |
|----|-------|-------|---------|-------|
| 1  | High  | Yes   | Yes     | Yes   |
| 2  | None  | No    | No      | No    |
| 3  | High  | Yes   | No      | Yes   |
| 4  | Mild  | No    | No      | No    |
| 5  | High  | No    | Yes     | Yes   |
| 6  | None  | Yes   | No      | No    |
| 7  | Mild  | Yes   | Yes     | Yes   |
| 8  | None  | No    | No      | No    |

**New Patient:** Fever=High, Cough=No, Fatigue=Yes

**Trees Vote:** Yes, Yes, No, Yes, Yes

### Result

4 say Sick, 1 says Healthy → **SICK** (Majority)



# Regression Example 1: House Price

**Dataset:** House details and selling prices

| Size(sqft) | Bedrooms | Age(yrs) | Location | Price  |
|------------|----------|----------|----------|--------|
| 1500       | 3        | 10       | Urban    | \$300K |
| 2000       | 4        | 5        | Suburb   | \$450K |
| 1200       | 2        | 20       | Rural    | \$180K |
| 1800       | 3        | 8        | Urban    | \$350K |
| 2500       | 4        | 2        | Suburb   | \$520K |
| 1400       | 3        | 15       | Urban    | \$280K |

**New House:** Size=1600, Beds=3, Age=12, Urban

**Trees Predict Different Prices:**

Tree 1: \$320K    Tree 2: \$340K    Tree 3: \$295K

Tree 4: \$350K    Tree 5: \$315K

**Average Price = \$324,000**

## Regression Example 2: Temperature Prediction

**Dataset:** Weather conditions and temperature

| Month | Humidity | Cloud% | Wind(km/h) | Temp(°C) |
|-------|----------|--------|------------|----------|
| Jan   | 60%      | 80     | 15         | 12       |
| Mar   | 45%      | 30     | 10         | 22       |
| Jun   | 70%      | 60     | 8          | 30       |
| Apr   | 50%      | 40     | 12         | 25       |
| Jul   | 75%      | 50     | 5          | 33       |
| Feb   | 55%      | 70     | 20         | 15       |

**New Day:** Month=May, Humidity=55%, Cloud=45, Wind=10

**Each Tree Predicts:**

Tree 1: 26°C    Tree 2: 24°C    Tree 3: 27°C

Tree 4: 25°C    Tree 5: 23°C

**Average = 25°C**

# Key Idea 1: Bagging (Different Data)

## Bootstrap = Random Sampling with Replacement

- Take data randomly (some rows repeat, some left out)
- Each tree trains on slightly different data

**Example:** Original 10 rows

| Sample   | Rows Selected                   |
|----------|---------------------------------|
| Sample 1 | [1, 3, 3, 5, 7, 2, 9, 4, 8, 10] |
| Sample 2 | [2, 4, 6, 8, 1, 3, 7, 9, 5, 5]  |
| Sample 3 | [5, 1, 9, 3, 7, 2, 8, 6, 4, 5]  |

- Row 3 appears twice in Sample 1
- Row 6 not in Sample 1 (used for testing!)

## Key Idea 2: Random Features

### Each tree looks at different features

- Have many features (Age, Income, Student, etc.)
- Each split: randomly pick only few features
- Tree chooses best among those few

### Example:

| Tree   | Features Considered      |
|--------|--------------------------|
| Tree 1 | Age, Income, Student     |
| Tree 2 | Age, Credit, Job         |
| Tree 3 | Income, Credit, Location |
| Tree 4 | Student, Job, Location   |

### Why?

Different features = Trees think differently = Better together!

# Classification: Categories (Majority Vote)

**Output:** Category (Spam/Not, Yes/No, Sick/Healthy)

**Each tree votes for a category → Majority wins!**

**Examples:**

- **Spam Detection:** 7 trees → 5 say Spam, 2 say Not → **Spam**
- **Customer Buy:** 5 trees → 4 say No, 1 says Yes → **No**
- **Disease:** 9 trees → 7 say Sick, 2 say Healthy → **Sick**

## Simple Rule

Count votes for each category, pick the one with most votes!

# Regression: Numbers (Average)

**Output:** Any number (Price, Temperature, Age)

**Each tree predicts a number → Average them all!**

## Examples:

- **House Price:** Trees predict [320K, 340K, 295K, 350K, 315K]
- Average = **\$324,000**
- **Temperature:** Trees predict [26, 24, 27, 25, 23]
- Average = **25°C**

## Simple Rule

Add all predictions, divide by number of trees!

## Building the Forest (Training):

- ① Start with your data
- ② For each tree (build 100 trees):
  - Take random sample of data
  - At each split, randomly pick few features
  - Choose best feature to split on
  - Grow tree completely

## Making Predictions (New Data):

- **For Categories:** Each tree votes → Majority wins
- **For Numbers:** Each tree predicts → Take average

# Real World Applications

## Category Prediction:

- **Email:** Spam vs Not Spam
- **Medical:** Sick vs Healthy
- **Banking:** Fraud vs Safe
- **Shopping:** Buy vs Not Buy
- **Loan:** Approve vs Reject
- **Face ID:** Match vs No Match

## Number Prediction:

- **Real Estate:** House price
- **Finance:** Stock price
- **Weather:** Temperature
- **Retail:** Sales forecast
- **Car:** Fuel efficiency
- **HR:** Salary prediction

## Bonus: Feature Importance

Forest also tells which features matter most!

- House price: Size(40%), Location(30%), Bedrooms(20%), Age(10%)



## Random Forest in One Line

Many diverse trees working together = Better predictions

### Remember:

- **Many Trees:** 100 trees better than 1 tree
- **Bagging:** Each tree sees different data
- **Random Features:** Each tree thinks differently
- **Categories:** Count votes, majority wins
- **Numbers:** Calculate average of all trees

Forest = Many Trees + Different Data + Different Feature

# Thank You!

Prepared by: Ananda Rimal

Questions?